

Opinion Perception Survey Analysis (preregistration)

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Invalid Date

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```
load_or_install <- function(pkg) {
  if (!require(pkg, character.only = TRUE)) {
    install.packages(pkg)
    library(pkg, character.only = TRUE)
  }
}

pkgs <- c(
  "tidyverse", "lme4", "lmerTest", "nloptr", "httpgd",
  "languageserver", "lmtest", "sandwich", "caret", "knitr",
  "prettycode", "performance", "estimatr", "patchwork"
)

knitr::opts_chunk$set(comment = NA)
invisible(lapply(pkgs, load_or_install))
options(vsc.viewer = TRUE)
theme(
  text = element_text(family = "Noto Sans")
)
```

```
)
options(tibble.print_max = Inf)
options(tibble.width = Inf)
```

Hypotheses and analysis plan

All hypotheses concerning a single slope parameter (e.g., H1 to H3) are tested using a t-test at significance level $\alpha = 0.05$. Hypotheses involving multiple parameters are tested using a single F-test on their joint significance, also at $\alpha = 0.05$. All subsequent hypotheses follow this convention without further repetition.

```
# -- Standardised test-result printers -----

#' One-sided t-test result block (for single beta)
print_one_sided <- function(label, beta, se, t, df = Inf,
                           direction = c("negative", "positive"),
                           ci = NULL) {
  direction <- match.arg(direction)
  p_two <- 2 * pt(-abs(t), df = df)
  p_one <- if ((direction == "negative" && beta < 0) ||
              (direction == "positive" && beta > 0)) p_two / 2 else 1 - p_two / 2
  supported <- p_one < 0.05 && ((direction == "negative" && beta < 0) ||
                                (direction == "positive" && beta > 0))

  cat(sprintf("=== %s ===\n", label))
  cat(sprintf("  Estimate   = %8.4f\n", beta))
  cat(sprintf("    SE        = %8.4f\n", se))
  cat(sprintf("    t         = %8.4f\n", t))
  cat(sprintf("    df         = %s\n", ifelse(is.finite(df), sprintf("%.1f", df), "infty"))))
  if (!is.null(ci))
    cat(sprintf("  95%% CI      = [%.4f, %.4f]\n", ci[1], ci[2]))
  cat(sprintf("    p (one-sided, beta %s 0) = %.4f\n",
              ifelse(direction == "negative", "<", ">"), p_one))
  cat(sprintf("  Supported : %s\n\n", ifelse(supported, "YES", "NO")))
  invisible(list(beta = beta, t = t, p_one = p_one, supported = supported))
}

#' LRT / F-test result block
print_lrt <- function(label, stat, df_num, df_den = NULL, p,
                      type = c("chi2", "F")) {
  type <- match.arg(type)
  cat(sprintf("=== %s ===\n", label))
  if (type == "chi2")
    cat(sprintf("  chi^2(%d) = %.3f,  p = %.4f  %s\n\n",
                df_num, stat, p, sig_stars(p)))
  else
    cat(sprintf("  F(%d, %d) = %.3f,  p = %.4f  %s\n\n",
                df_num, df_den, stat, p, sig_stars(p)))
  invisible(list(stat = stat, p = p))
}

#' Model-fit comparison table
print_fit_table <- function(...) {
  models <- list(...)
  tbl <- data.frame(
    Model    = names(models),
    AIC      = round(sapply(models, AIC), 1),
    BIC      = round(sapply(models, BIC), 1),
    logLik   = round(sapply(models, \(m) as.numeric(logLik(m))), 1),
    npar     = sapply(models, \(m) length(fixef(m)))
  )
  tbl$delta_AIC <- tbl$AIC - min(tbl$AIC)
  tbl$delta_BIC <- tbl$BIC - min(tbl$BIC)
  print(tbl, row.names = FALSE)
  invisible(tbl)
```

```

}

sig_stars <- function(p) {
  case_when(p < .001 ~ "***", p < .01 ~ "**", p < .05 ~ "*",
            p < .1 ~ ".", TRUE ~ "ns")
}

# Extract lmer beta, SE, t, Satterthwaite df in one call
lmer_coef <- function(model, term) {
  cf <- summary(model)$coefficients
  list(
    beta = cf[term, "Estimate"],
    se   = cf[term, "Std. Error"],
    t    = cf[term, "t value"],
    df   = cf[term, "df"],           # added by lmerTest
    p2   = cf[term, "Pr(>|t|)"]
  )
}

```

```

questions_sc <- c(
  "climate_concern", "gay_marriage", "rights_indep_integration",
  "econ_inequality", "regulate_internet", "east_germans"
)

parties      <- c("Left Party", "BSW", "Green Party", "SPD", "FDP", "CDU/CSU",
                  "AfD", "No party", "Other party", "Refuse to say/No answer")
partiesVars  <- c("LeftParty", "BSW", "GreenParty", "SPD", "FDP", "CDU/CSU", "AfD")
partiesCloseSel <- c("Left Party", "BSW", "Green Party", "SPD", "FDP", "CDU/CSU", "AfD")

# Shared column-name vectors (used across many hypotheses)
col_deltas   <- paste0("deltaX_",      questions_sc)
col_alpha_exp <- paste0("exp_alpha_",  questions_sc)
col_alpha_lin <- paste0("linear_alpha_", questions_sc)
col_weights  <- paste0("w_",          questions_sc)
col_var_std  <- paste0("std_socialCircle_ops_", questions_sc)
col_var_var  <- paste0("var_socialCircle_ops_", questions_sc)

```

2 Load Data

```

df_p <- read.csv(
  "processed_data/2026-06-19_data_processed_participant_withAllIssueWeights.csv"
)

inds_bothwaves <- df_p |>
  count(id) |>
  filter(n == 2) |>
  pull(id)

treatment_ids <- df_p |>
  filter(wave == 2, treatment_wave2 == TRUE, id %in% inds_bothwaves) |>
  pull(id)
control_ids <- df_p |>
  filter(wave == 2, treatment_wave2 == FALSE, id %in% inds_bothwaves) |>
  pull(id)

df_p <- df_p |>
  mutate(
    overall_std_socialCircle = rowMeans(across(all_of(col_var_std))),
    across(all_of(col_var_std), \(x) x^2,
           .names = "var_{gsub('std_', '', col)}"),
    wave = factor(wave)
  )

df_diff <- read.csv(

```

```
"processed_data/2026-06-19_data_processed_differences_withAllIssueWeights.csv"
) |>
mutate(
  ingroupdummy = as.integer(ingroupdummy == "True"),
  wave         = factor(wave)
) |>
left_join(
  df_p |> select(id, wave,
                 all_of(paste0("var_socialCircle_ops_", questions_sc)),
                 all_of(paste0("std_socialCircle_ops_", questions_sc))),
  by = c("id", "wave")
)
```

3 Hypotheses

3.1 Aim 1

Aim 1: Compare approaches to measuring perceived political differences

3.1.1 H1

3.1.1.1 H1. Participants place the typical voter of the party they affiliate with nearer to themselves than other typical voters.

We model the distance between the dots representing the participant (self) and the typical voter of each party r as

$$d_{n,\text{self},r} = \beta_0 + \beta_1 \mathbb{I}(r = \text{party}_n) + u_n + \epsilon_{n,r}, \quad (1)$$

where $\mathbb{I}(\cdot)$ is an indicator function, u_n is a participant level random intercept, and $\epsilon_{n,r}$ is an error term.

The prediction is $\beta_1 < 0$. Participants who prefer a political party place the typical voter of that party closer to themselves than the average of the typical voters of other parties.

```
df_h1 <- df_diff |>
  filter(wave %in% c(1, 2), dot1 == "self", dot2 %in% partiesVars) |>
  select(id, wave, ingroupdummy, party, pixel_dist)

m_h1 <- lmer(pixel_dist ~ ingroupdummy + wave + (1 | id), data = df_h1, REML = TRUE)
ci_h1 <- confint(m_h1, parm = "ingroupdummy", method = "Wald")
cf_h1 <- lmer_coef(m_h1, "ingroupdummy")

print_one_sided("H1: Ingroup party placed closer to self",
  cf_h1$beta, cf_h1$se, cf_h1$t, cf_h1$df,
  direction = "negative", ci = ci_h1)
```

=== H1: Ingroup party placed closer to self ===

```
Estimate = -0.2520
SE        = 0.0041
t         = -61.5081
df        = 13785.6
95% CI    = [-0.2600, -0.2440]
p (one-sided, beta < 0) = 0.0000
Supported : YES
```

3.1.2 H2

3.1.2.1 H2. Distances to voters on the map correspond to likeability ratings.

We model likeability ratings of the typical voter of political party r (rated only in wave 2) as a function of map distances:

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + u_n + \epsilon_{n,r}. \quad (2)$$

The prediction is $\beta_1 < 0$. Voters placed farther away from the participant's own dot on the map receive lower likeability ratings than voters placed closer to the participant's own dot.

```
df_h2 <- df_diff |>
  filter(wave == 2, dot1 == "self", dot2 %in% partiesVars) |>
  select(id, sympathy, party, pixel_dist)
```

```

m_h2 <- lmer(sympathy ~ pixel_dist + (1 | id), data = df_h2, REML = TRUE)
ci_h2 <- confint(m_h2, parm = "pixel_dist", method = "Wald")
cf_h2 <- lmer_coef(m_h2, "pixel_dist")

print_one_sided("H2: Map distance predicts likeability",
                cf_h2$beta, cf_h2$se, cf_h2$t, cf_h2$df,
                direction = "negative", ci = ci_h2)

```

```

=== H2: Map distance predicts likeability ===
Estimate  = -1.1108
SE         =  0.0160
t          = -69.6313
df         = 6723.9
95% CI     = [-1.1421, -1.0795]
p (one-sided, beta < 0) = 0.0000
Supported : YES

```

3.1.3 H3

3.1.3.1 H3. Distances on the political map correspond to similarity ratings.

Similar to H2, we model similarity ratings between a pair of individuals, i and j , as a function of map distances:

$$s_{n,ij} = \beta_0 + \beta_1 \cdot d_{n,ij} + u_n + \epsilon_{n,ij} . \quad (3)$$

The prediction is $\beta_1 < 0$. Pairs placed farther apart on the map receive lower similarity ratings than pairs placed closer together.

```

df_h3 <- df_diff |>
  filter(wave %in% c(1, 2)) |>
  select(id, wave, pairwise_similarity, pixel_dist) |>
  drop_na()

m_h3 <- lmer(pairwise_similarity ~ pixel_dist + wave + (1 | id), data = df_h3, REML = TRUE)
ci_h3 <- confint(m_h3, parm = "pixel_dist", method = "Wald")
cf_h3 <- lmer_coef(m_h3, "pixel_dist")

print_one_sided("H3: Map distance predicts pairwise similarity",
                cf_h3$beta, cf_h3$se, cf_h3$t, cf_h3$df,
                direction = "negative", ci = ci_h3)

```

```

=== H3: Map distance predicts pairwise similarity ===
Estimate  = -0.7761
SE         =  0.0066
t          = -117.4882
df         = 38233.2
95% CI     = [-0.7890, -0.7631]
p (one-sided, beta < 0) = 0.0000
Supported : YES

```

3.1.4 H4

3.1.4.1 H4. The closest pair on the map is rated most similar and the farthest pair is rated least similar

Consider $(ij)^{min}$ the pair of individuals i and j with the smallest distance on participant n 's political map and $(ij)^{max}$ the pair with the largest distance on the map. The similarity difference between these two pairs

$$\Delta s_{max} = s_{n,(ij)^{min}} - s_{n,(ij)^{max}} . \quad (4)$$

The prediction is $\Delta s_{max} > 0$ (standard t-test with $\alpha = 0.05$). The closest pair on the map receives a higher similarity rating than the farthest pair.

```

df_h4_wide <- df_diff |>
  filter(wave %in% c(1, 2),
         minPixelDistance_pair_dummy == "True" | maxPixelDistance_pair_dummy == "True") |>
  select(id, wave, maxPixelDistance_pair_dummy, pairwise_similarity) |>
  pivot_wider(
    id_cols = c(id, wave),

```

```

names_from = maxPixelDistance_pair_dummy,
values_from = pairwise_similarity
) |>
rename(sim_min = "False", sim_max = "True") |>
mutate(diff = sim_min - sim_max)

```

```
t_h4 <- t.test(df_h4_wide$diff, mu = 0)
```

```

cat("=== H4: Closest pair rated most similar ===\n")
cat(sprintf("  Mean(sim_min) = %.4f ± %.4f\n", mean(df_h4_wide$sim_min), sd(df_h4_wide$sim_min)))
cat(sprintf("  Mean(sim_max) = %.4f ± %.4f\n", mean(df_h4_wide$sim_max), sd(df_h4_wide$sim_max)))
cat(sprintf("  Mean diff      = %.4f\n",          mean(df_h4_wide$diff)))
cat(sprintf("  t              = %.4f\n",          t_h4$statistic))
cat(sprintf("  p (one-sided) = %.4f\n",          t_h4$p.value / 2))
cat(sprintf("  Supported      : %s\n\n",
            ifelse(t_h4$statistic > 0 && t_h4$p.value / 2 < 0.05, "YES", "NO")))

```

```

=== H4: Closest pair rated most similar ===
  Mean(sim_min) = 0.7113 ± 0.2688
  Mean(sim_max) = 0.2308 ± 0.2771
  Mean diff      = 0.4805
  t              = 55.3777
  p (one-sided) = 0.0000
  Supported      : YES

```

3.1.5 H5

3.1.5.1 H5. The inferred issue weights are robust across waves.

We test the reliability of our tool by comparing the similarity of issue weights, $\alpha_{n,q}$, of participants in wave 1 vs.~wave 2 (control condition) with the similarity of issue weights between participants.

$$\overline{\Delta\alpha_{q_{\text{within}}}} = \mathbb{E} [|\alpha_{n,q}(\text{wave 2}) - \alpha_{n,q}(\text{wave 1})|] \quad \text{and} \quad (5)$$

$$\overline{\Delta\alpha_{q_{\text{between}}}} = \mathbb{E} [|\alpha_{n,q}(\text{wave 2}) - \alpha_{m,q}(\text{wave 1/wave 2})|] \quad \text{with } m \neq n \quad (6)$$

ERROR: It should be: $\overline{\Delta\alpha_{q_{\text{within}}}} - \overline{\Delta\alpha_{q_{\text{between}}}} < 0$

The prediction is that $\overline{\Delta\alpha_{q_{\text{within}}}} - \overline{\Delta\alpha_{q_{\text{between}}}} > 0$ (one-sample t-test with $\alpha = 0.05$). Participants' issue weights in wave 2 are more similar to their own issue weights in the previous wave than to those of other participants in wave 1 or wave 2 (in the control condition).

```

k <- "exp"
alpha_cols <- paste0(k, "_alpha_", questions_sc)

ctrl_ids_h5 <- df_p |>
  filter(id %in% inds_bothwaves, wave == 2, treatment_wave2 == "0") |>
  pull(id)

df_h5 <- df_p |>
  filter(if_all(all_of(alpha_cols), Negate(is.na)), id %in% ctrl_ids_h5) |>
  select(id, wave, all_of(alpha_cols))

ids_both <- intersect(
  df_h5 |> filter(wave == 1) |> pull(id),
  df_h5 |> filter(wave == 2) |> pull(id)
)
df_h5 <- df_h5 |> filter(id %in% ids_both)

w1_mat <- df_h5 |> filter(wave == 1) |> arrange(id) |> select(all_of(alpha_cols)) |> as.matrix()
w2_mat <- df_h5 |> filter(wave == 2) |> arrange(id) |> select(all_of(alpha_cols)) |> as.matrix()
n <- nrow(w1_mat)

within_means <- rowMeans(abs(w2_mat - w1_mat))
between_means <- vapply(seq_len(n), \(i) {
  mean(rowMeans(abs(sweep(w2_mat[-i, , drop = FALSE], 2, w1_mat[i, ]))))
}, numeric(1))

```

```

delta      <- within_means - between_means
t_h5       <- t.test(delta, mu = 0, alternative = "two.sided")
cohens_d   <- mean(delta) / sd(delta)

cat("=== H5: Issue weights reliable across waves ===\n")
cat(sprintf("  Within (mean ± sd) = %.4f ± %.4f\n", mean(within_means), sd(within_means)))
cat(sprintf("  Between (mean ± sd) = %.4f ± %.4f\n", mean(between_means), sd(between_means)))
cat(sprintf("  Delta (mean)      = %.4f\n", mean(delta)))
cat(sprintf("  Cohen's d          = %.4f\n", cohens_d))
cat(sprintf("  t                  = %.4f\n", t_h5$statistic))
cat(sprintf("  p (two-sided)       = %.4f\n", t_h5$p.value))
cat(sprintf("  p (one-sided)      = %.4f\n", t_h5$p.value / 2))
cat(sprintf("  Supported          : %s\n",
            ifelse(t_h5$statistic < 0 && t_h5$p.value / 2 < 0.05, "YES", "NO")))

```

```

=== H5: Issue weights reliable across waves ===
  Within (mean ± sd) = 0.1988 ± 0.0777
  Between (mean ± sd) = 0.2211 ± 0.0259
  Delta (mean)      = -0.0223
  Cohen's d          = -0.3094
  t                  = -6.1494
  p (two-sided)       = 0.0000
  p (one-sided)      = 0.0000
  Supported          : YES

```

3.1.6 Exploration

3.1.6.1 Explorative Analyses.

In addition to the preregistered hypotheses above, we will explore how people rate the political mapping task against the pairwise similarity task in terms of difficulty and enjoyment, how satisfied they are with the political map, and how much time they needed for each task.

```
# (see 03_data-overview)
```

3.2 Aim 2

Aim 2: Identify patterns linked to (political) identity

In this section, we hypothesise that demographic and identity-related variables, including party preference, $party_n$, left-right placement, lr_n , age, age_n , or combinations of them moderate the relationship between belief differences and issue weights.

3.2.1 H6

3.2.1.1 H6. Party preference predicts issue weights.

We model the (normalised) issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \sum_r \beta_{r,q} \cdot \mathbb{I}[\text{party}_n = r] + u_q + \epsilon_{n,q} \quad (7)$$

where $\mathbb{I}[\text{party}_n = r]$ are dummy variables for each party r (with one party as reference category), $\beta_{r,q}$ is the effect of party r on weight assigned to issue q , u_q is the average deviation of the weight for issue q , and $\epsilon_{n,q}$ is the individual- and issue-specific error.

The prediction is $\beta_{r,q} \neq 0$ and $\beta_{r1,q} \neq \beta_{r2,q}$ for at least some parties $r1$ and $r2$. Participants affiliated with different parties differ in how strongly they weigh different political issues when mapping individuals.

3.2.1.2 Interpretation 1: Do parties differ across issues?

For correlations (unconstrained):

```

# Helper: run H6 for one weight type and print verdict
run_h6 <- function(k, use_lmer = TRUE) {
  alpha_cols <- paste0(k, "_alpha_", questions_sc)

  df_h6 <- df_p |>
    filter(wave %in% c(1, 2)) |>
    select(id, wave, party_close, all_of(alpha_cols)) |>
    filter(party_close %in% c(partiesCloseSel, "No party")) |>
    pivot_longer(

```

```

      cols      = all_of(alpha_cols),
      names_to   = "issue",
      names_prefix = paste0(k, "_alpha_"),
      values_to   = "alpha_q"
    ) |>
  filter(!is.na(party_close), !is.na(alpha_q)) |>
  mutate(
    party_close = relevel(factor(party_close), ref = "No party"),
    issue       = relevel(factor(issue),       ref = "climate_concern"),
    wave        = factor(wave)
  )

if (use_lmer) {
  m_null <- lmer(alpha_q ~ issue + wave + (1 | id),          data = df_h6)
  m_full <- lmer(alpha_q ~ party_close * issue + wave + (1 | id), data = df_h6)
  wt      <- anova(m_null, m_full)
  stat    <- wt$Chisq[2]; df_num <- wt$Df[2]; p <- wt$`Pr(>Chisq)`[2]
  print_fit_table("Null" = m_null, "Full" = m_full)
  print_lrt(sprintf("H6 (%s): party × issue interaction", k),
    stat, df_num, p = p, type = "chi2")
} else {
  m_null <- lm(alpha_q ~ issue + wave,          data = df_h6)
  m_full <- lm(alpha_q ~ party_close * issue + wave, data = df_h6)
  wt      <- anova(m_null, m_full)
  stat    <- wt$F[2]; df_num <- wt$Df[2]; df_den <- wt$Res.Df[2]; p <- wt$`Pr(>F)`[2]
  # clustered-SE model for coefficient display
  m_full_rob <- lm_robust(alpha_q ~ party_close * issue + wave,
    data = df_h6, clusters = id, se_type = "CR2")
  cat(sprintf("\nR^2 Null = %.3f | Full = %.3f\n",
    summary(m_null)$r.squared, summary(m_full)$r.squared))
  print_lrt(sprintf("H6 (%s): party × issue interaction", k),
    stat, df_num, df_den, p, type = "F")
}
}

run_h6("corrS", use_lmer = TRUE)

```

refitting model(s) with ML (instead of REML)

```
run_h6("exp", use_lmer = FALSE)
```

```

Model      AIC      BIC logLik npar delta_AIC delta_BIC
Null -3880.3 -3813.5 1949.2   7      48      0.0
Full -3928.3 -3549.7 2015.1  49      0     263.8

```

```

=== H6 (corrS): party × issue interaction ===
chi^2(42) = 397.031, p = 0.0000 ***

```

```
R^2 Null = 0.008 | Full = 0.014
```

```

=== H6 (exp): party × issue interaction ===
F(42, 12335) = 1.701, p = 0.0031 **

```

And for exponential/linear weights (constrained):

```

# Per-issue breakdown (both weight types)
run_h6_per_issue <- function(k, use_lmer = TRUE) {
  cat(sprintf("\n-- H6 per issue (%s) --\n", k))

  purrr::walk(questions_sc, \(q) {
    alpha_col <- paste0(k, "_alpha_", q)
    df_q <- df_p |>
      filter(wave %in% c(1, 2)) |>
      select(id, wave, party_close, alpha_q = all_of(alpha_col)) |>
      filter(party_close %in% c(partiesCloseSel, "No party"), !is.na(alpha_q)) |>
      mutate(party_close = relevel(factor(party_close), ref = "No party"),
        wave             = factor(wave))
  })
}

```



```

if (use_lmer) {
  m      <- lmer(alpha_q ~ party_close + wave + (1 | id), data = df_q)
  m_null <- lmer(alpha_q ~ wave + (1 | id),               data = df_q)
  wt     <- anova(m_null, m)
  p      <- wt$`Pr(>Chisq)`[2]; chi <- wt$Chisq[2]; df_lrt <- wt$Df[2]
  r2v    <- r2(m)
  cat(sprintf("\n%s  R^2=%.3f/%.3f  LRT chi^2(%d)=%.2f  p=%.4f  %s\n",
              q, r2v$R2_marginal, r2v$R2_conditional,
              df_lrt, chi, p, sig_stars(p)))
} else {
  m      <- lm_robust(alpha_q ~ party_close + wave, data = df_q,
                      clusters = id, se_type = "CR2")
  m_lm   <- lm(alpha_q ~ party_close + wave, data = df_q)
  m_null <- lm(alpha_q ~ wave,               data = df_q)
  wt     <- anova(m_null, m_lm)
  p      <- wt$`Pr(>F)`[2]; f <- wt$F[2]
  cat(sprintf("\n%s  R^2=%.3f  F(%d,%d)=%.2f  p=%.4f  %s\n",
              q, summary(m_lm)$r.squared,
              wt$Df[2], wt$Res.Df[2], f, p, sig_stars(p)))
}
})
}

run_h6_per_issue("corrS", use_lmer = TRUE)

```

```

refitting model(s) with ML (instead of REML)
refitting model(s) with ML (instead of REML)
refitting model(s) with ML (instead of REML)
refitting model(s) with ML (instead of REML)
refitting model(s) with ML (instead of REML)
refitting model(s) with ML (instead of REML)

```

```
run_h6_per_issue("exp", use_lmer = FALSE)
```

```
-- H6 per issue (corrS) --
```

```
climate_concern  R^2=0.066/0.566  LRT chi^2(7)=103.31  p=0.0000 ***
```

```
gay_marriage     R^2=0.075/0.529  LRT chi^2(7)=120.27  p=0.0000 ***
```

```
rights_indep_integration  R^2=0.049/0.542  LRT chi^2(7)=77.51  p=0.0000 ***
```

```
econ_inequality  R^2=0.035/0.480  LRT chi^2(7)=56.93  p=0.0000 ***
```

```
regulate_internet  R^2=0.023/0.413  LRT chi^2(7)=39.52  p=0.0000 ***
```

```
east_germans      R^2=0.014/0.436  LRT chi^2(7)=22.49  p=0.0021 **
```

```
-- H6 per issue (exp) --
```

```
climate_concern  R^2=0.006  F(7,2055)=1.80  p=0.0831 .
```

```
gay_marriage     R^2=0.006  F(7,2055)=1.37  p=0.2119 ns
```

```
rights_indep_integration  R^2=0.007  F(7,2055)=1.94  p=0.0600 .
```

```
econ_inequality  R^2=0.005  F(7,2055)=1.56  p=0.1444 ns
```

```
regulate_internet  R^2=0.003  F(7,2055)=0.99  p=0.4333 ns
```

```
east_germans      R^2=0.009  F(7,2055)=2.34  p=0.0222 *
```

3.2.2 H7

3.2.2.1 H7. Left-Right placement predicts issue weights.

We model the issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot lr_n + u_q + \epsilon_{n,q} \tag{8}$$

with continuous variable lr_n between left' (\$-5\$) and right' (5). We estimate this model jointly across all six issues. % beta_0 is average. Because alpha is normalised: beta_0 = 1/6 % lr_n is explanatory variable % u_q is average deviation of the weight for issue q % eps_{nq} is individual specific error for this question and this individual.

The prediction is $\beta_q \neq 0$. Left and right participants differ in how strongly they weigh different political issues when mapping individuals. %We test this via a single F-test on the joint significance of $\{\beta_q\}$, with significance threshold $\alpha = 0.05$.

```
k <- "exp"

cat("=== H7: Left-right placement predicts issue weights ===\n\n")

# Per-issue
purrr::walk(questions_sc, \(q) {
  df_q <- df_p |>
    filter(wave %in% c(1, 2)) |>
    select(id, wave, lr, alpha_q = all_of(paste0(k, "_alpha_", q))) |>
    filter(!is.na(alpha_q))
  m <- lmer(alpha_q ~ lr + wave + (1 | id), data = df_q)
  cf <- lmer_coef(m, "lr")
  cat(sprintf(" %-35s beta=%.4f t=%.3f p=%.4f %s\n",
              q, cf$beta, cf$t, cf$p2, sig_stars(cf$p2)))
})

# Joint model (LRT)
df_long_h7 <- df_p |>
  filter(wave %in% c(1, 2)) |>
  select(id, wave, lr, starts_with(paste0(k, "_alpha_"))) |>
  pivot_longer(starts_with(paste0(k, "_alpha_")),
               names_to = "issue", values_to = "alpha") |>
  mutate(issue = gsub(paste0(k, "_alpha_"), "", issue), issue = factor(issue)) |>
  filter(!is.na(alpha))

m_full_h7 <- lmer(alpha ~ issue + issue:lr + wave + (1 | id), data = df_long_h7)

boundary (singular) fit: see help('isSingular')

m_null_h7 <- lmer(alpha ~ issue + wave + (1 | id), data = df_long_h7)

boundary (singular) fit: see help('isSingular')

wt_h7 <- anova(m_null_h7, m_full_h7)

refitting model(s) with ML (instead of REML)

cat("\n")
print_lrt("H7 joint: lr x issue",
          wt_h7$Chisq[2], wt_h7$Df[2], p = wt_h7$`Pr(>Chisq)`[2], type = "chi2")

=== H7: Left-right placement predicts issue weights ===

climate_concern          beta=-0.0528  t=-2.163  p=0.0307  *
gay_marriage              beta=-0.0515  t=-2.144  p=0.0322  *
rights_indep_integration  beta=0.0173  t=0.681  p=0.4958  ns
econ_inequality           beta=0.0349  t=1.477  p=0.1400  ns
regulate_internet         beta=0.0225  t=1.088  p=0.2768  ns
east_germans              beta=0.0285  t=1.182  p=0.2374  ns

=== H7 joint: lr x issue ===
chi^2(6) = 16.826,  p = 0.0099  **
```

3.2.3 H8

3.2.3.1 H8. Age predicts issue weights.

We model the issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot age_n + u_q + \epsilon_{n,q} \tag{9}$$

The prediction is $\beta_q \neq 0$. Older and younger participants differ in how strongly they weigh different political issues when mapping individuals.

```
k <- "exp"

cat("=== H8: Age predicts issue weights ===\n\n")

purrr::walk(questions_sc, \(q) {
  df_q <- df_p |>
    filter(wave %in% c(1, 2)) |>
    select(id, wave, age, alpha_q = all_of(paste0(k, "_alpha_", q))) |>
    filter(!is.na(alpha_q), !is.na(age)) |>
    mutate(age_z = as.numeric(scale(age)))
  m <- lmer(alpha_q ~ age_z + wave + (1 | id), data = df_q, REML = FALSE)
  cf <- lmer_coef(m, "age_z")
  cat(sprintf("  %-35s beta=%.4f t=%.3f p=%.4f %s\n",
              q, cf$beta, cf$t, cf$p2, sig_stars(cf$p2)))
})

# Joint model
df_long_h8 <- df_p |>
  filter(wave %in% c(1, 2), !is.na(age)) |>
  mutate(age_z = as.numeric(scale(age))) |>
  select(id, wave, age_z, starts_with(paste0(k, "_alpha_"))) |>
  pivot_longer(starts_with(paste0(k, "_alpha_")),
               names_to = "issue", values_to = "alpha") |>
  mutate(issue = gsub(paste0(k, "_alpha_"), "", issue), issue = factor(issue)) |>
  filter(!is.na(alpha))

m_full_h8 <- lmer(alpha ~ issue + issue:age_z + wave + (1 | id), data = df_long_h8, REML = FALSE)

boundary (singular) fit: see help('isSingular')

m_null_h8 <- lmer(alpha ~ issue + wave + (1 | id),
                  data = df_long_h8, REML = FALSE)

boundary (singular) fit: see help('isSingular')

wt_h8      <- anova(m_null_h8, m_full_h8)

cat("\n")
print_lrt("H8 joint: age x issue",
          wt_h8$Chisq[2], wt_h8$Df[2], p = wt_h8$`Pr(>Chisq)`[2], type = "chi2")

=== H8: Age predicts issue weights ===

  climate_concern      beta=0.0144  t=2.780  p=0.0055  **
  gay_marriage         beta=-0.0070  t=-1.377  p=0.1689  ns
  rights_indep_integ    beta=-0.0084  t=-1.571  p=0.1164  ns
  econ_inequality      beta=-0.0074  t=-1.468  p=0.1425  ns
  regulate_internet    beta=0.0051  t=1.156  p=0.2480  ns
  east_germans         beta=0.0034  t=0.671  p=0.5024  ns

=== H8 joint: age x issue ===
  chi^2(6) = 18.230,  p = 0.0057  **

3.2.4 Exploration
3.2.4.1 Explorative Analyses.

We also model H6 to H8 with different combinations of party preference, left-right placement, and age of the participants to model participants issue weights  $\alpha_{n,q}$ . In addition, we model how political interest,  $pi_n$ ,  $pf_n$ , interacts with the explanatory variables above.

print(" exploration later ")

[1] " exploration later "
```

3.3 Aim 3

Aim 3: Assess the role of social circle belief variation

For Aim 3, we test whether including social circle belief variation improves predictions for participants’ issue weights (inferred from the map distances), their reported issue importance, likeability of voters, and polarisation perceptions. In addition, we explore how social circle belief improves predictions of the map distances between individuals.

3.3.1 H9

3.3.1.1 H9. Social circle variance predicts issue weights.

We model issue weights of participant n as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot \text{var}_{n,q} + u_q + \epsilon_{n,q} \cdot \tag{10}$$

The prediction is $\beta_q < 0$. Issues are assigned higher weight when associated with lower social circle variance.

CHANGE TO PREREG: I am using standard deviation instead of variance now!!!

```
run_h9 <- function(k = "exp", variance_type = "std") {
  var_col <- paste0(variance_type, "_socialCircle_ops")

  df_h9 <- df_p |>
    filter(wave %in% c(1, 2)) |>
    pivot_longer(
      cols          = matches(paste0("^(", var_col, "|", k, "_alpha)_")),
      names_to      = c(".value", "issue"),
      names_pattern = paste0("^(", var_col, "|", k, "_alpha)_(.*)"")
    ) |>
    rename(var_sc = all_of(var_col), alpha = all_of(paste0(k, "_alpha"))) |>
    filter(!is.na(alpha), !is.na(var_sc))

  m <- lmer(alpha ~ var_sc + wave + (1 | issue), data = df_h9)
  cf <- lmer_coef(m, "var_sc")
  print_one_sided(
    sprintf("H9: Social circle %s predicts issue weights (%s)", variance_type, k),
    cf$beta, cf$se, cf$t, direction = "negative"
  )

  # Per-issue
  cat("Per-issue breakdown:\n")
  purrr::walk(questions_sc, \(q) {
    df_q <- df_p |>
      filter(wave %in% c(1, 2)) |>
      select(id, wave,
        var_sc = all_of(paste0(var_col, "_", q)),
        alpha  = all_of(paste0(k, "_alpha_", q))) |>
      filter(!is.na(alpha), !is.na(var_sc))
    m_q <- lmer(alpha ~ var_sc + wave + (1 | id), data = df_q)
    cf_q <- lmer_coef(m_q, "var_sc")
    cat(sprintf("  %-35s beta=%7.4f t=%6.3f p=%.4f %s\n",
      q, cf_q$beta, cf_q$t, cf_q$p2, sig_stars(cf_q$p2)))
  })
  cat("\n")
}
```

```
run_h9("exp", "std")
run_h9("corrS", "std")
```

```
=== H9: Social circle std predicts issue weights (exp) ===
Estimate  = -0.0788
SE        =  0.0098
t         = -8.0041
df        = infty
p (one-sided, beta < 0) = 0.0000
Supported : YES
```

Per-issue breakdown:

climate_concern	beta=-0.0668	t=-2.659	p=0.0079	**
gay_marriage	beta=-0.0807	t=-3.665	p=0.0003	***
rights_indep_integration	beta=-0.0161	t=-0.648	p=0.5169	ns
econ_inequality	beta=-0.0506	t=-1.866	p=0.0622	.
regulate_internet	beta=-0.1390	t=-6.158	p=0.0000	***
east_germans	beta=-0.1574	t=-5.752	p=0.0000	***

=== H9: Social circle std predicts issue weights (corrS) ===

```
Estimate = -0.1095
SE        = 0.0108
t         = -10.1444
df        = infty
p (one-sided, beta < 0) = 0.0000
Supported : YES
```

Per-issue breakdown:

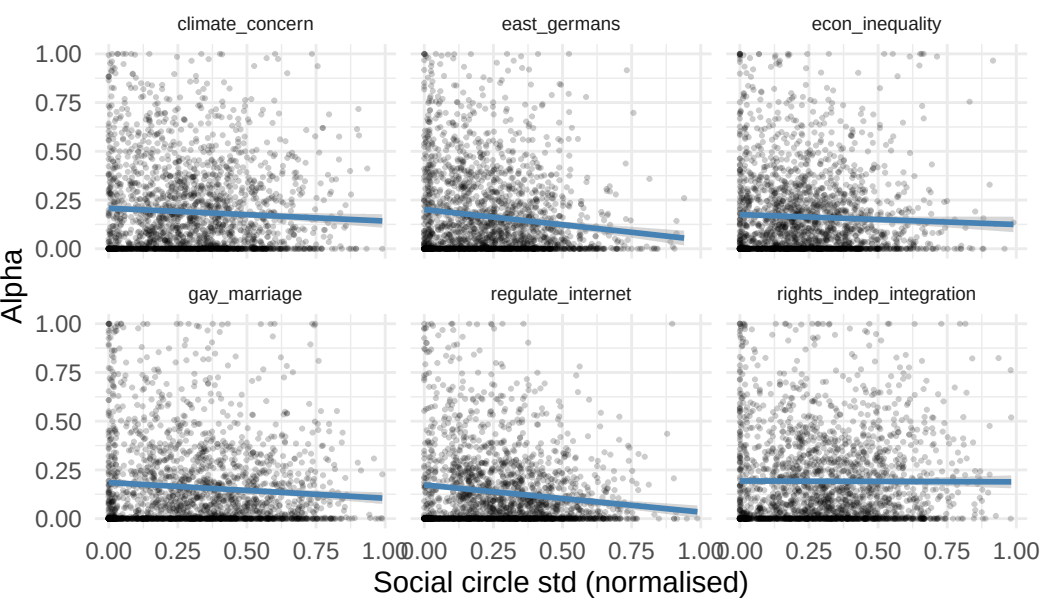
climate_concern	beta=-0.0647	t=-2.343	p=0.0192	*
gay_marriage	beta=-0.1176	t=-4.778	p=0.0000	***
rights_indep_integration	beta=-0.0546	t=-2.056	p=0.0399	*
econ_inequality	beta=-0.1530	t=-5.190	p=0.0000	***
regulate_internet	beta=-0.1965	t=-7.353	p=0.0000	***
east_germans	beta=-0.2255	t=-8.318	p=0.0000	***

```
k <- "exp"; variance_type <- "std"
df_h9_plot <- df_p |>
  filter(wave %in% c(1, 2)) |>
  pivot_longer(
    cols      = matches(paste0("^std_socialCircle_ops|", k, "_alpha_")),
    names_to   = c(".value", "issue"),
    names_pattern = paste0("^std_socialCircle_ops|", k, "_alpha_(.*)")
  ) |>
  rename(var_sc = std_socialCircle_ops, alpha = all_of(paste0(k, "_alpha"))) |>
  filter(!is.na(alpha), !is.na(var_sc))

ggplot(df_h9_plot, aes(x = var_sc, y = alpha)) +
  geom_point(alpha = 0.2, size = 0.4) +
  geom_smooth(method = "lm", se = TRUE, color = "steelblue") +
  facet_wrap(~ issue, ncol = 3) +
  labs(title = sprintf("H9: Social circle %s vs %s alpha", variance_type, k),
       x = sprintf("Social circle %s (normalised)", variance_type), y = "Alpha") +
  theme_minimal() +
  theme(strip.text = element_text(size = 7))
```

`geom\_smooth()` using formula = 'y ~ x'

H9: Social circle std vs exp alpha



```
ggsave(sprintf("h9_plot_%s_%s.png", k, variance_type), width = 16/2.54, height = 9/2.54)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

3.3.2 H10

3.3.2.1 H10. Social circle variance predicts reported issue importance.

We model reported issue importance of participant n as

$$w_{n,q} = \beta_0 + \beta_q \cdot \text{var}_{n,q} + u_q + v_n + \epsilon_{n,q} . \tag{11}$$

where v_n is the average importance of all issues assigned by participant n .

The prediction is $\beta_q < 0$. Issues are rated as more important when social circle variance is smaller.

```
variance_type <- "std"
var_col        <- paste0(variance_type, "_socialCircle_ops")

df_h10 <- df_p |>
  filter(wave %in% c(1, 2)) |>
  pivot_longer(
    cols        = c(all_of(paste0("w_", questions_sc)),
                     all_of(paste0(var_col, "_", questions_sc))),
    names_to     = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, "|w)_(.*)" )
  ) |>
  rename(var_sc = all_of(var_col))

m_h10 <- lmer(w ~ var_sc + wave + (1 | issue) + (1 | id), data = df_h10)
cf_h10 <- lmer_coef(m_h10, "var_sc")

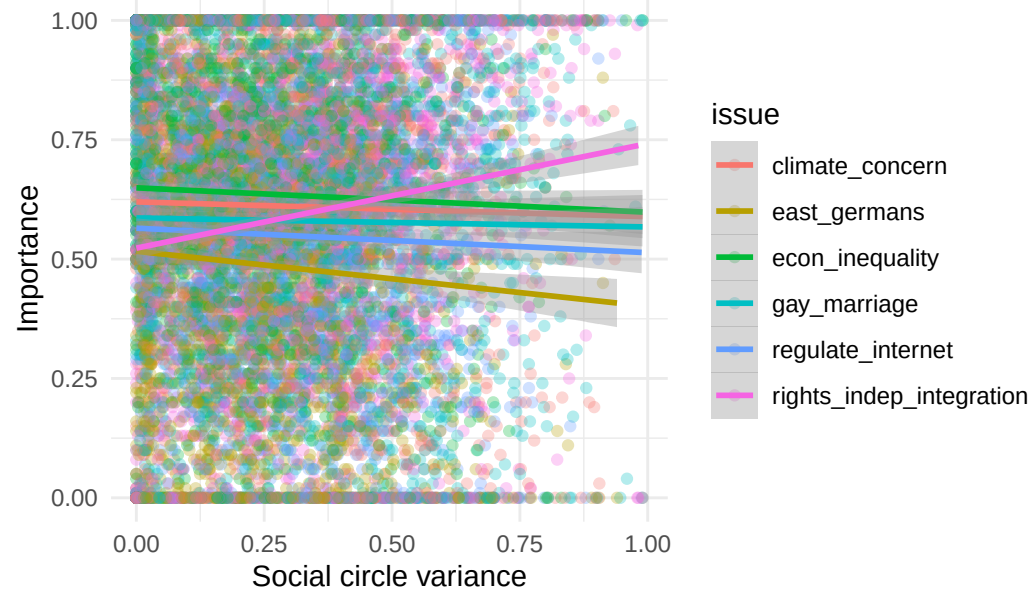
print_one_sided(
  sprintf("H10: Social circle %s predicts reported issue importance", variance_type),
  cf_h10$beta, cf_h10$se, cf_h10$t, direction = "negative"
)

# Per-issue
cat("Per-issue breakdown:\n")
purrr::walk(questions_sc, \(q) {
  df_q <- df_h10 |> filter(issue == q) |> select(id, wave, issue, w, var_sc)
  m_q   <- lmer(w ~ var_sc + wave + (1 | id), data = df_q)
  cf_q  <- lmer_coef(m_q, "var_sc")
  cat(sprintf(" %-35s beta=%7.4f t=%6.3f p=%.4f %s\n",
              q, cf_q$beta, cf_q$t, cf_q$p2, sig_stars(cf_q$p2)))
})

ggplot(df_h10, aes(x = var_sc, y = w, color = issue)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = "Social circle variance", y = "Importance",
       title = sprintf("H10: Social circle %s vs relative importance", variance_type)) +
  theme_minimal()

`geom_smooth()` using formula = 'y ~ x'
```

H10: Social circle std vs relative importance



=== H10: Social circle std predicts reported issue importance ===

```
Estimate = 0.0053
SE       = 0.0141
t        = 0.3748
df       = inf
p (one-sided, beta < 0) = 0.6461
Supported : NO
```

Per-issue breakdown:

climate_concern	beta=-0.0079	t=-0.260	p=0.7952	ns
gay_marriage	beta=-0.0176	t=-0.589	p=0.5559	ns
rights_indep_integration	beta= 0.1817	t= 5.917	p=0.0000	***
econ_inequality	beta=-0.0584	t=-1.919	p=0.0551	.
regulate_internet	beta=-0.0528	t=-1.704	p=0.0885	.
east_germans	beta=-0.1060	t=-3.146	p=0.0017	**

3.3.3 H11

3.3.3.1 H11. Overall social circle variance helps predict likeability rating of typical voters.

In H2 (equation 2), we modelled the participant n 's likeability rating of the typical voter of political party r as

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + u_n + \epsilon_{n,r}. \quad (12)$$

Here, we extend this model as

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + \beta_2 \cdot \sum_q \text{var}_{n,q} + v_n + \epsilon_{n,r} \quad (13)$$

where the new β_2 -term represents the influence of overall social circle belief variation on likeability.

The prediction is $\beta_2 > 0$ and that equation 14 improves the predictions of the baseline model from **H2** (equation 2/12) using R2, AIC, and BIC as model quality indicators. Voters are more liked in general if the participant's overall social circle variance is higher.

TODO: This hypothesis is kind of wrong. We have 1 avgsocialcirclestd per person and 7 pixeldist and sympathy for 7 parties. But the avgsocialcirclestd is already included in the random intercept.

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + \beta_2 \cdot \sum_q \text{var}_{n,q} \text{REMOVED}(+v_n) + \epsilon_{n,r} \quad (14)$$

```
variance_type <- "std"

avg_var_sc <- df_p |>
  filter(wave == 2) |>
  select(id, all_of(paste0(variance_type, "_socialCircle_ops_", questions_sc))) |>
  drop_na() |>
  mutate(avg_var_sc = rowMeans(across(everything(), .names = NULL))) |>
  select(id, avg_var_sc)
```



```

df_h11 <- df_diff |>
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, sympathy, party, pixel_dist) |>
  left_join(avg_var_sc, by = "id") |>
  mutate(pixel_dist_z = as.numeric(scale(pixel_dist)),
         avg_var_sc_z = as.numeric(scale(avg_var_sc)))

m_base_h11 <- lmer(sympathy ~ pixel_dist_z + (1 | id),
                  data = df_h11, REML = FALSE)
m_ext_h11 <- lmer(sympathy ~ pixel_dist_z + avg_var_sc_z + (1 | id),
                  data = df_h11, REML = FALSE)

cf_h11 <- lmer_coef(m_ext_h11, "avg_var_sc_z")

# One-sided test: H1 = avg_var_sc_z has a positive effect
p_one_h11 <- if (cf_h11$beta > 0) {
  cf_h11$p2 / 2
} else {
  1 - cf_h11$p2 / 2
}

r2_base <- r2(m_base_h11)
r2_ext <- r2(m_ext_h11)

cat("=== H11: Social circle variance predicts likeability (beyond map distance) ===\n")
cat(sprintf(" beta (pixel_dist) = %.4f\n",
            fixef(m_ext_h11)["pixel_dist_z"]))
cat(sprintf(" beta (avg_var_sc) = %.4f\n",
            cf_h11$beta))
cat(sprintf(" SE = %.4f\n",
            cf_h11$se))
cat(sprintf(" t = %.4f\n",
            cf_h11$t))
cat(sprintf(" df (Satterthwaite) = %.2f\n",
            cf_h11$df))
cat(sprintf(" p (two-sided) = %.6f\n",
            cf_h11$p2))
cat(sprintf(" p (one-sided, beta > 0) = %.6f\n",
            p_one_h11))
cat(sprintf(" Supported: %s\n",
            ifelse(p_one_h11 < 0.05 && cf_h11$beta > 0,
                  "YES", "NO")))
cat(sprintf(" R² marginal base/ext = %.4f / %.4f (Delta=%+.4f)\n",
            r2_base$R2_marginal,
            r2_ext$R2_marginal,
            r2_ext$R2_marginal - r2_base$R2_marginal))
cat(sprintf(" R² conditional base/ext = %.4f / %.4f (Delta=%+.4f)\n",
            r2_base$R2_conditional,
            r2_ext$R2_conditional,
            r2_ext$R2_conditional - r2_base$R2_conditional))
cat(sprintf(" AIC base/ext = %.2f / %.2f (Delta=%+.2f)\n",
            AIC(m_base_h11),
            AIC(m_ext_h11),
            AIC(m_ext_h11) - AIC(m_base_h11)))
cat(sprintf(" BIC base/ext = %.2f / %.2f (Delta=%+.2f)\n\n",
            BIC(m_base_h11),
            BIC(m_ext_h11),
            BIC(m_ext_h11) - BIC(m_base_h11)))

```

```

=== H11: Social circle variance predicts likeability (beyond map distance) ===
beta (pixel_dist) = -0.2258
beta (avg_var_sc) = -0.0070
SE = 0.0052
t = -1.3357

```



```
df (Satterthwaite) = 849.13
p (two-sided) = 0.181989
p (one-sided, beta > 0) = 0.909006
Supported: NO
R² marginal base/ext = 0.4208 / 0.4214 (Delta=+0.0006)
R² conditional base/ext = 0.5802 / 0.5783 (Delta=-0.0019)
AIC base/ext = 313.82 / 308.70 (Delta=-5.13)
BIC base/ext = 341.08 / 342.76 (Delta=+1.68)
```

3.3.4 H12a

3.3.4.1 H12a. Social circle variance predicts perceived political polarisation on a given issue.

We model perceived polarisation on issue q as

$$P_{n,q} = \beta_0 + \beta_q \cdot var_{n,q} + u_n + \epsilon_{n,q}. \quad (15)$$

The prediction is $\beta_q < 0$. Participants with less variation in their social circles with respect to issue q perceive higher polarisation on that issue. %We test this via a single F-test on the joint significance of $\{\beta_q\}$, with significance threshold $\alpha = 0.05$.

```
variance_type <- "std"
var_col       <- paste0(variance_type, "_socialCircle_ops")

df_h12a <- df_p |>
  filter(wave %in% c(1, 2)) |>
  select(-P_tot) |>
  pivot_longer(
    cols      = c(all_of(paste0("P_", questions_sc)),
                  all_of(paste0(var_col, "_", questions_sc))),
    names_to   = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, "|P)_(.*)" ) |>
  rename(var_sc = all_of(var_col))

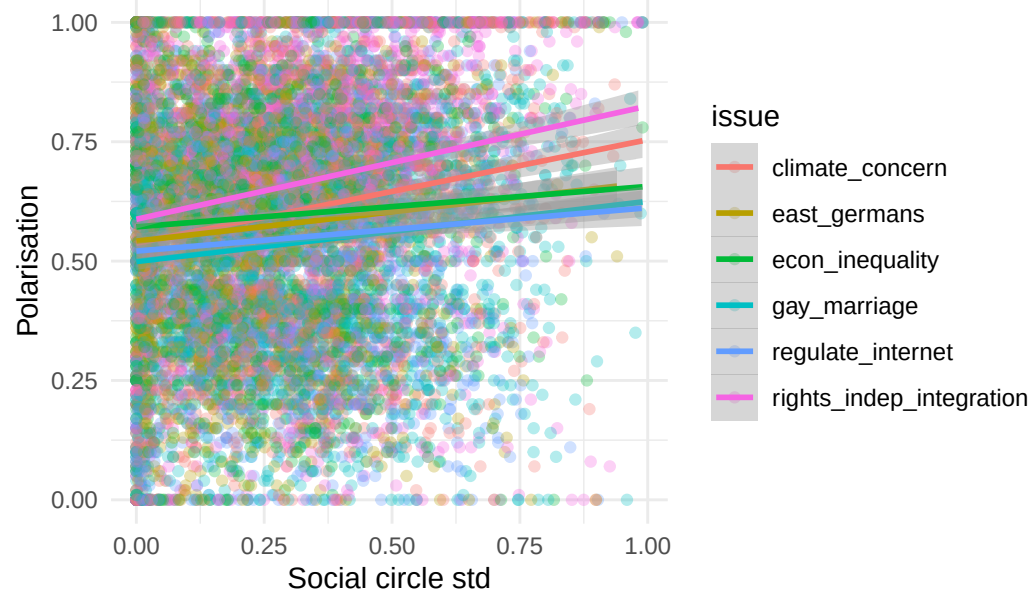
m_h12a <- lmer(P ~ var_sc + wave + (1 | issue) + (1 | id), data = df_h12a)
cf_h12a <- lmer_coef(m_h12a, "var_sc")

print_one_sided(
  sprintf("H12a: Social circle %s predicts perceived polarisation (per issue)", variance_type),
  cf_h12a$beta, cf_h12a$se, cf_h12a$t, direction = "negative"
)

ggplot(df_h12a, aes(x = var_sc, y = P, color = issue)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = sprintf("Social circle %s", variance_type), y = "Polarisation",
       title = sprintf("H12a: Social circle %s vs issue polarisation", variance_type)) +
  theme_minimal()

`geom_smooth()` using formula = 'y ~ x'
```

H12a: Social circle std vs issue polarisation



=== H12a: Social circle std predicts perceived polarisation (per issue) ===

```
Estimate  =  0.1685
SE        =  0.0117
t         = 14.4028
df        = inf
p (one-sided, beta < 0) = 1.0000
Supported : NO
```

3.3.5 H12b

3.3.5.1 H12b. Social circle variance predicts perceived political polarisation.

We model overall perceived polarisation as

$$P_{n,tot} = \beta_0 + \beta_1 \cdot \sum_q var_{n,q} + \epsilon_n. \quad (16)$$

The prediction is \$ \beta_1 < 0 \$. Participants with less variation in their social circles averaged over all issues perceive more political polarisation. %We test this via a t-test with significance threshold $\alpha = 0.05$.

```
df_h12b <- df_p |>
  select(id, wave, all_of(paste0(variance_type, "_socialCircle_ops_", questions_sc)), P_tot) |>
  drop_na() |>
  mutate(avg_var_sc = rowSums(across(all_of(paste0(variance_type, "_socialCircle_ops_", questions_sc)))))

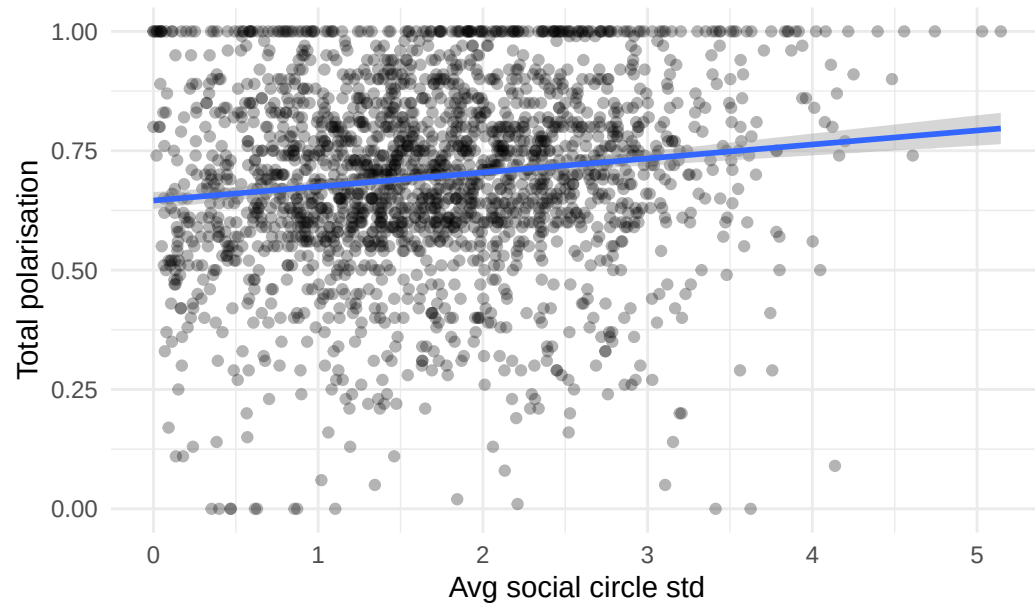
m_h12b <- lmer(P_tot ~ avg_var_sc + (1 | id), data = df_h12b, REML = TRUE)
n_p_h12b <- n_distinct(df_h12b$id)
cf_h12b <- lmer_coef(m_h12b, "avg_var_sc")
t_h12b <- cf_h12b$beta / cf_h12b$se

print_one_sided(
  "H12b: Overall social circle variance predicts total perceived polarisation",
  cf_h12b$beta, cf_h12b$se, t_h12b, df = n_p_h12b - 2,
  direction = "negative"
)

ggplot(df_h12b, aes(x = avg_var_sc, y = P_tot)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = sprintf("Avg social circle %s", variance_type), y = "Total polarisation",
       title = sprintf("H12b: Social circle %s vs total polarisation", variance_type)) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

H12b: Social circle std vs total polarisation



=== H12b: Overall social circle variance predicts total perceived polarisation ===

```
Estimate  = 0.0234
SE        = 0.0050
t         = 4.6493
df        = 1225.0
p (one-sided, beta < 0) = 1.0000
Supported : NO
```

3.3.6 H13

3.3.6.1 H13. Reported issue importance predicts perceived distance.

In equation ??, we fix the issue weights to $\alpha_{n,q} = w_{n,q}$. Then, we model the perceived distance between two individuals i and j (using the exponential kernel; see equations ?? and ??) as

$$d_{n,ij} = 1 - \exp \left(-\hat{b}_n \cdot \left(\sum_q w_{n,q} \cdot |x_{n,i,q} - x_{n,j,q}| \right)^{\hat{c}_n} \right) \quad (17)$$

with fit parameters $\hat{b}_n$, and $\hat{c}_n$.

The prediction is that this model (with $\hat{\alpha}_{n,q} = w_{n,q}$) outperforms (i) an unweighted model with $\hat{\alpha}_{n,q} = 1/6$ and (ii) a population weight model with $\hat{\alpha}_{n,q} = 1/n \cdot \sum_n w_{n,q}$. We use R2, AIC, and BIC as model quality indicators.

```
linear_kernel <- function(x, a, b) a + b * x
exp_kernel    <- function(x, a, b) 1 - exp(-a * (x ^ b))
r2_vec        <- function(y, pm) {
  ss_tot <- sum((y - mean(y, na.rm = TRUE))^2, na.rm = TRUE)
  1 - colSums((pm - y)^2, na.rm = TRUE) / ss_tot
}
compute_aic_bic <- function(mse, k, n) {
  rss <- mse * n
  tibble(aic = n * log(rss / n) + 2 * k, bic = n * log(rss / n) + k * log(n))
}

alphas_pop_lin <- df_p |> select(all_of(col_alpha_lin)) |> colMeans()
alphas_pop_exp <- df_p |> select(all_of(col_alpha_exp)) |> colMeans()

df_h13_base <- df_diff |>
  filter(wave %in% c(1, 2)) |>
  left_join(df_p |> select(id, wave, all_of(col_weights),
                          all_of(col_alpha_lin), all_of(col_alpha_exp)),
            by = c("id", "wave"), suffix = c("", ".drop")) |>
  select(-ends_with(".drop"))

.d <- as.matrix(df_h13_base[, col_deltas])
```

```

.a_uniform <- matrix(1 / length(questions_sc), nrow = nrow(df_h13_base), ncol = length(questions_sc))
.a_pop_exp <- matrix(alphas_pop_exp, nrow = nrow(df_h13_base), ncol = length(questions_sc), byrow = TRUE)
.a_pop_lin <- matrix(alphas_pop_lin, nrow = nrow(df_h13_base), ncol = length(questions_sc), byrow = TRUE)
.a_fit_exp <- as.matrix(df_h13_base[, col_alpha_exp])
.a_fit_lin <- as.matrix(df_h13_base[, col_alpha_lin])
.a_weights <- {
  w <- as.matrix(df_h13_base[, col_weights])
  w_sum <- rowSums(w, na.rm = TRUE)
  ifelse(w_sum == 0, 0, w / w_sum)
}

```

```

pred_cols <- c("predicted_d_weights", "predicted_d_population_exp",
              "predicted_d_population_lin", "predicted_d_uniform_exp",
              "predicted_d_uniform_lin", "predicted_d_fit_linear",
              "predicted_d_fit_exp")

```

```

df_h13 <- df_h13_base |>
  mutate(
    predicted_d_uniform_exp = exp_kernel( rowSums(.d * .a_uniform), exp_param1, exp_param2),
    predicted_d_uniform_lin = linear_kernel(rowSums(.d * .a_uniform), linear_param1, linear_param2),
    predicted_d_population_exp = exp_kernel( rowSums(.d * .a_pop_exp), exp_param1, exp_param2),
    predicted_d_population_lin = linear_kernel(rowSums(.d * .a_pop_lin), linear_param1, linear_param2),
    predicted_d_fit_exp = exp_kernel( rowSums(.d * .a_fit_exp), exp_param1, exp_param2),
    predicted_d_fit_linear = linear_kernel(rowSums(.d * .a_fit_lin), linear_param1, linear_param2),
    predicted_d_weights = exp_kernel( rowSums(.d * .a_weights), exp_param1, exp_param2)
  ) |>
  mutate(across(all_of(pred_cols),
    ~ pixel_dist - .x,
    .names = "err_{str_remove(.col, 'predicted_d_')}"))

```

```

y <- df_h13$pixel_dist
n_obs <- sum(!is.na(y))
n_pw <- df_h13 |> distinct(id, wave) |> nrow()
n_q <- length(questions_sc)
pred_mat <- df_h13 |> select(all_of(pred_cols)) |> as.matrix()
colnames(pred_mat) <- str_remove(pred_cols, "predicted_d_")

```

```

k_lookup <- tribble(
  ~model, ~k,
  "weights", 2,
  "population_exp", 2 + n_q,
  "population_lin", 2 + n_q,
  "uniform_exp", 2,
  "uniform_lin", 2,
  "fit_exp", (2 + n_q) * n_pw,
  "fit_linear", (2 + n_q) * n_pw
)

```

```

baseline_sq <- (pred_mat[, "uniform_exp"] - y)^2

```

```

comparison_tbl <- tibble(
  model = colnames(pred_mat),
  mse = colMeans((pred_mat - y)^2, na.rm = TRUE),
  rmse = sqrt(mse),
  r2 = r2_vec(y, pred_mat)
) |>
  mutate(
    rel_improve = (mse[model == "uniform_exp"] - mse) / mse[model == "uniform_exp"],
    p_wilcox = map_dbl(model, \(m) {
      if (m == "uniform_exp") return(NA_real_)
      wilcox.test((pred_mat[, m] - y)^2, baseline_sq, paired = TRUE)$p.value
    }),
    mean_diff_vs_uniform = map_dbl(model, \(m) {
      if (m == "uniform_exp") return(NA_real_)
      mean((pred_mat[, m] - y)^2 - baseline_sq, na.rm = TRUE)
    })
  )

```

```

) |>
left_join(k_lookup, by = "model") |>
mutate(
  aic_bic = map2(mse, k, ~ compute_aic_bic(.x, .y, n_obs)),
  aic      = map_dbl(aic_bic, "aic"),
  bic      = map_dbl(aic_bic, "bic")
) |>
select(-aic_bic, -k) |>
arrange(mse)

print(comparison_tbl)

```

```

# A tibble: 7 x 9
  model      mse  rmse      r2 rel_improve p_wilcox
<chr>    <dbl> <dbl>    <dbl>    <dbl>    <dbl>
1 fit_linear 0.0162 0.127  0.509      0.237      0
2 fit_exp    0.0164 0.128  0.505      0.230      0
3 population_exp 0.0212 0.146  0.358      0.00238 2.54e-71
4 uniform_exp 0.0213 0.146  0.357      0          NA
5 weights    0.0330 0.182  0.00175    -0.552      0
6 uniform_lin 0.0431 0.208 -0.301     -1.02      4.13e-14
7 population_lin 0.0466 0.216 -0.407     -1.19      1.29e- 2

  mean_diff_vs_uniform      aic      bic
      <dbl>    <dbl>    <dbl>
1    -0.00505  -1102767.  -923289.
2    -0.00490  -1100181.  -920702.
3    -0.0000506 -1062758. -1062674.
4          NA    -1062113. -1062092.
5     0.0117    -940918.  -940897.
6     0.0218    -867738.  -867717.
7     0.0253    -846120.  -846036.

```

3.3.6.2 Analysis

```

err_cols <- str_replace(pred_cols, "predicted_d_", "err_")

# Per-participant R^2
per_person_r2 <- df_h13 |>
  group_by(id) |>
  summarise(across(all_of(pred_cols), \(x) {
    v <- !is.na(x) & !is.na(pixel_dist)
    if (sum(v) < 3) return(NA_real_)
    ss_res <- sum((pixel_dist[v] - x[v])^2)
    ss_tot <- sum((pixel_dist[v] - mean(pixel_dist[v]))^2)
    if (ss_tot == 0) NA_real_ else 1 - ss_res / ss_tot
  }, .names = "r2_{.col}")) |>
  pivot_longer(starts_with("r2_"), names_to = "model", values_to = "r2") |>
  mutate(model = str_remove(model, "r2_predicted_d_"))

p1 <- ggplot(per_person_r2, aes(x = reorder(model, r2, median, na.rm = TRUE), y = r2)) +
  geom_boxplot(outlier.alpha = 0.8, outlier.size = 0.2) +
  stat_summary(fun = mean, geom = "point", shape = 15, size = 3, colour = "red") +
  geom_hline(yintercept = 0, colour = "red", linetype = "dashed") +
  coord_flip() + ylim(-2, 1) +
  labs(title = "H13: Per-participant R^2", x = NULL, y = expression(R^2)) +
  theme_minimal()

# Predicted vs observed
p2 <- df_h13 |>
  slice_sample(n = 10000) |>
  select(pixel_dist, all_of(pred_cols)) |>
  pivot_longer(-pixel_dist, names_to = "model", values_to = "predicted") |>
  mutate(model = str_remove(model, "predicted_d_")) |>
  ggplot(aes(pixel_dist, predicted)) +
  geom_point(alpha = 0.05, size = 0.3) +

```

```
geom_abline(slope = 1, intercept = 0, colour = "red") +
geom_smooth(method = "gam", formula = y ~ s(x, bs = "cs"), se = FALSE, colour = "steelblue") +
facet_wrap(~ model, ncol = 4) + ylim(0, 1) + coord_equal() +
labs(title = "H13: Predicted vs observed") + theme_minimal()
```

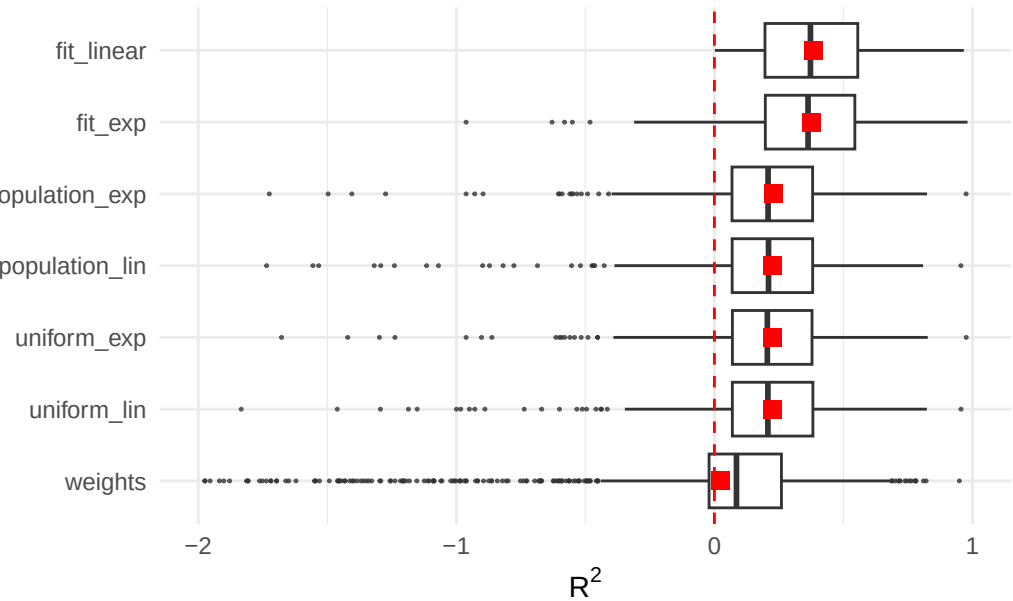
```
# Participant-level MSE
p3 <- df_h13 |>
  group_by(id) |>
  summarise(across(all_of(str_replace(pred_cols, "predicted_d_", "err_")),
    \ (x) mean(x^2, na.rm = TRUE))) |>
  pivot_longer(-id, names_to = "model", values_to = "mse") |>
  mutate(model = str_remove(model, "^err_")) |>
  ggplot(aes(x = reorder(model, mse, median), y = mse)) +
  geom_boxplot(outlier.alpha = 0.15) +
  coord_flip() + ylim(0, 0.1) +
  labs(title = "H13: Participant-level MSE", x = NULL, y = "MSE") +
  theme_minimal()
```

```
p1; p2; p3
```

```
Warning: Removed 94 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

```
Warning: Removed 94 rows containing non-finite outside the scale range
(`stat_summary()`).
```

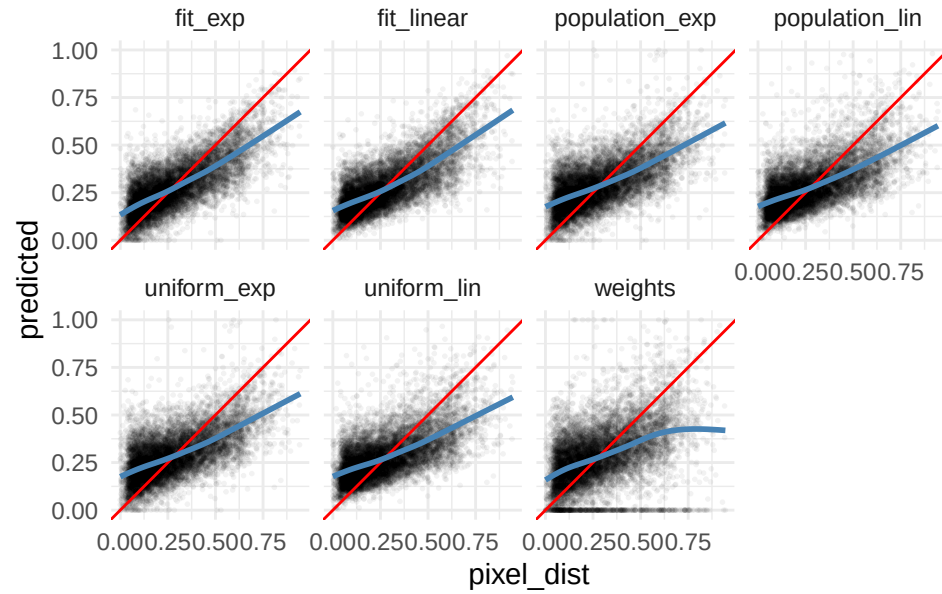
H13: Per-participant R^2



```
Warning: Removed 77 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

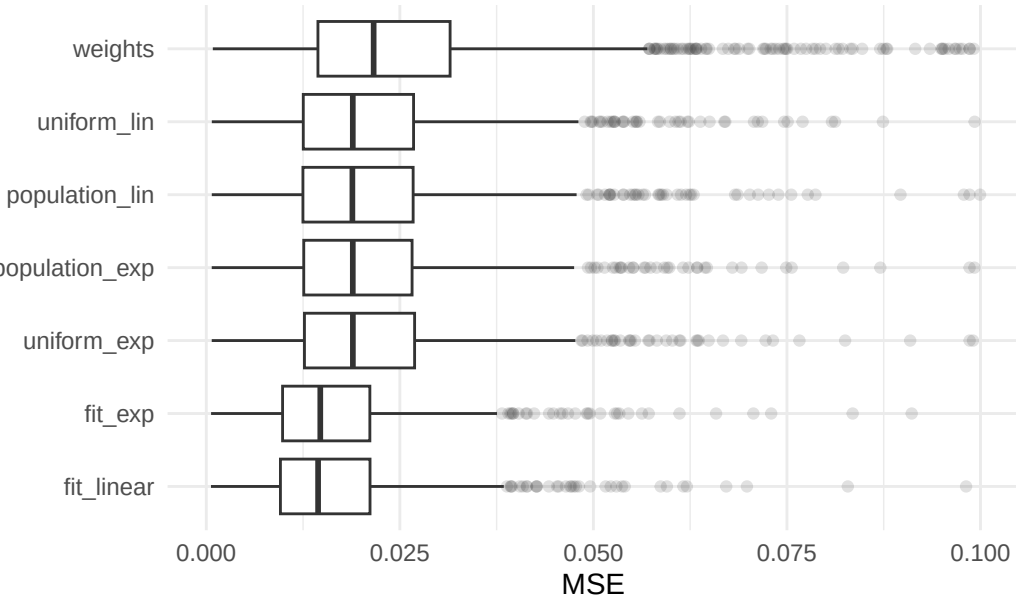
```
Warning: Removed 77 rows containing missing values or values outside the scale range
(`geom_point()`).
```

H13: Predicted vs observed



Warning: Removed 115 rows containing non-finite outside the scale range (``stat_boxplot()``).

H13: Participant-level MSE



3.3.6.3 Explorative Analyses.

In addition to the preregistered hypotheses above, we explore several ways to define the function g_n (equation ??) which describes the (participant-level) aggregation of belief differences on the six items q between individuals i and j into an overall belief distance.

In particular, we model different functional forms of g_n based on theories about who consider as social context and how this social context shapes their perception of the belief space—which we refer to as a social *lens*.

`%\begin{mybox}{Exploration of different lens models}` We define three different functional types of aggregation functions g_n with increasing level of complexity. First, a **simple lens** $\mathcal{S}_n$ is a 6-dimensional vector that scales (or weighs) each entry in the belief distance vector (equivalent to the weighted metric g_n in Box 1). Second, a **compression lens** $\mathcal{C}_n$ is a 6-dimensional vector, which projects the belief difference vector onto a single dimension, e.g. `~liberal'` vs. `~conservative'`, and scales the belief distance in the direction of this dimension. Third, a **full lens** $\mathcal{F}_n$ is a 6×6 matrix that transforms the belief distance vector into a subjective new space with the same number of dimensions but with rotated and scaled axes. Writing these lenses in terms of transformation matrices yields:

- 1 simplified lens $g_n(|x_{n,i,q} - x_{n,j,q}|) = \left\| \begin{pmatrix} 0 & \\ & \mathcal{S}_n \end{pmatrix} \circ \begin{pmatrix} |x_{n,i,1} - x_{n,j,1}| \\ \vdots \\ |x_{n,i,6} - x_{n,j,6}| \end{pmatrix} \right\|_p$
- 2 compression lens $g_n(|x_{n,i,q} - x_{n,j,q}|) = \left\| \begin{pmatrix} - & \mathcal{C}_n & - \\ 0 & \dots & 0 \\ 0 & \vdots & 0 \end{pmatrix} \circ \begin{pmatrix} |x_{n,i,1} - x_{n,j,1}| \\ \vdots \\ |x_{n,i,6} - x_{n,j,6}| \end{pmatrix} \right\|_p$
- 3 full lens $g_n(|x_{n,i,q} - x_{n,j,q}|) = \left\| \begin{pmatrix} \mathcal{F}_{n,11} & \dots & \mathcal{F}_{n,61} \\ \vdots & \dots & \vdots \\ \mathcal{F}_{n,61} & \dots & \mathcal{F}_{n,66} \end{pmatrix} \circ \begin{pmatrix} |x_{n,i,1} - x_{n,j,1}| \\ \vdots \\ |x_{n,i,6} - x_{n,j,6}| \end{pmatrix} \right\|_p$

where $\|\cdot\|_p$ is the p-norm. Here, we choose Minkowski norm $p = 1$ or Euclidean norm $p = 2$.

The lenses $\mathcal{S}_n$, $\mathcal{C}_n$, and $\mathcal{F}_n$ of participant n depend on the belief distribution in the participant's social contexts, $x_{n,\text{context},q}$. We use two ways to define this social contexts:

- the beliefs of the seven to ten social circle contacts of the participant: $x_{n,\text{context},q} = \{x_{n,k,q} \mid k = 1 : K_n\}$.
- the beliefs of all the other participants in the participant's in-group, where in-group can mean (i) those who affiliate with the same party, $\{x_{m,\text{self},q} \mid m \text{ with party}_m = \text{party}_n\}$, (ii) those who place themselves in the same region on the left-right scale, (iii) those who have similar demographic attributes, or (iv) combinations thereof.

Then, we determine participant n 's lenses ...

- $\mathcal{S}_n$ as the inverse of the standard deviation of the $x_{n,\text{context},q}$. $\mathcal{S}_n = \left(\frac{1}{\text{Var}(x_{n,\text{context},1})}, \dots, \frac{1}{\text{Var}(x_{n,\text{context},6})}\right)$.
- $\mathcal{C}_n$ as the first (loaded) principal components of the social context beliefs. $\mathcal{C}_n = \frac{1}{\sqrt{\lambda_{n,1}}} \nu_{n,1}$.
- $\mathcal{F}_n$ as a principal component transformation, i.e as the inverse of the loaded principal component vectors of the social context beliefs. $\mathcal{F}_n = \left(\sqrt{\lambda_{n,1}} \nu_{n,1}, \dots, \sqrt{\lambda_{n,6}} \nu_{n,6}\right)^{-1}$, where $\nu_{n,\tilde{q}}$ are principal components and $\lambda_{n,\tilde{q}}$ the corresponding eigenvalues.

We predict map distances $d_{n,ij}$ using equation ?? with different choices of lenses and different definitions of social contexts. The goal of this exploration is to investigate to what extent these (theoretical) models and assumptions about participant-level social lenses describe how people perceive political differences between individuals.

3.4 Aim 4

Aim 4: Test the effects of an intervention on social circle belief variation

In wave 2, participants are randomly put in (i) a control condition, where they receive the same instruction as in wave 1, namely to choose seven to ten social contacts that they would count to their social circle, or (ii) a treatment condition, where they receive an additional instruction to choose contacts such that they represent a diverse mix of beliefs, if possible. Aim 4 is to test the effects of this instruction.

3.4.1 H14

3.4.1.1 H14. Participants in the treatment condition report higher social circle variance.

For the participants in the treatment condition T , who receive the instruction to choose diverse contacts, we measure the increase in social circle variance on issue q as

$$\Delta \text{var}_T = \sum_q (\mathbb{E}[\text{var}_{n,q}(w2) \mid \text{condition}_n = T] - \mathbb{E}[\text{var}_{n,q}(w1) \mid \text{condition}_n = T]) \tag{18}$$

The prediction is $\sim \Delta \text{var}_T > 0$ (one-sample t-test with significance threshold $\alpha = 0.05$) and $\Delta \text{var}_T > \Delta \text{var}_{\text{Control}}$ (using independent samples t-test with $\alpha = 0.05$). Social circle variance in the treatment condition T is higher than in wave 1 and higher than in the control group.

```
variance_type <- "std"
var_col       <- paste0(variance_type, "_socialCircle_ops")

df_h14 <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(
    cols      = matches(paste0(var_col, "_")),
    names_to   = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, ")_(.*)")
  ) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
              names_prefix = "var_sc_wave", values_from = all_of(var_col)) |>
  mutate(group = if_else(id %in% treatment_ids, "treatment", "control"),
         delta = var_sc_wave2 - var_sc_wave1)

cat("=== H14: Treatment increases social circle variance ===\n\n")

# Sanity check
df_h14 |>
  group_by(group) |>
  summarise(mean_w1 = mean(var_sc_wave1, na.rm = TRUE),
            sd_w1    = sd(var_sc_wave1,   na.rm = TRUE),
            mean_w2 = mean(var_sc_wave2, na.rm = TRUE),
            sd_w2    = sd(var_sc_wave2,   na.rm = TRUE),
            mean_d   = mean(delta,        na.rm = TRUE),
```

```
n = n()) |> print()
```

```
t_bal <- t.test(var_sc_wavel ~ group, data = df_h14)
cat(sprintf(" Balance check (wavel by group): t=%.3f p=%.4f\n\n", t_bal$statistic, t_bal$p.value))

trt <- df_h14 |> filter(group == "treatment")
ctrl <- df_h14 |> filter(group == "control")

# H14-1: Deltavar_T > 0
t_h14_1 <- t.test(trt$delta, mu = 0, alternative = "greater")
cat(sprintf("H14-1 (Deltavar_treatment > 0):\n"))
cat(sprintf(" Mean delta = %.4f ± %.4f\n", mean(trt$delta, na.rm = TRUE), sd(trt$delta, na.rm = TRUE)))
cat(sprintf(" t = %.4f, p (one-sided) = %.4f\n", t_h14_1$statistic, t_h14_1$p.value))
cat(sprintf(" Supported : %s\n\n", ifelse(t_h14_1$statistic > 0 && t_h14_1$p.value < 0.05, "YES", "NO")))

# H14-2: Deltavar_T > Deltavar_Control
t_h14_2 <- t.test(delta ~ group, data = df_h14, alternative = "less", var.equal = FALSE)
cat(sprintf("H14-2 (Deltavar_treatment > Deltavar_control):\n"))
cat(sprintf(" Mean delta treatment = %.4f ± %.4f\n", mean(trt$delta, na.rm = TRUE), sd(trt$delta, na.rm = TRUE)))
cat(sprintf(" Mean delta control = %.4f ± %.4f\n", mean(ctrl$delta, na.rm = TRUE), sd(ctrl$delta, na.rm = TRUE)))
cat(sprintf(" Difference (T - C) = %.4f\n", mean(trt$delta, na.rm = TRUE) - mean(ctrl$delta, na.rm = TRUE)))
cat(sprintf(" t = %.4f, p (one-sided) = %.4f\n", t_h14_2$statistic, t_h14_2$p.value))
cat(sprintf(" Supported : %s\n\n", ifelse(t_h14_2$statistic < 0 && t_h14_2$p.value < 0.05, "YES", "NO")))
```

=== H14: Treatment increases social circle variance ===

```
# A tibble: 2 x 7
  group mean_w1 sd_w1 mean_w2 sd_w2 mean_d n
<chr> <dbl> <dbl> <dbl> <dbl> <dbl> <int>
1 control 0.268 0.198 0.275 0.197 0.00638 2370
2 treatment 0.285 0.199 0.311 0.205 0.0262 3042
Balance check (wavel by group): t=-3.044 p=0.0023
```

```
H14-1 (Deltavar_treatment > 0):
Mean delta = 0.0262 ± 0.1884
t = 7.6679, p (one-sided) = 0.0000
Supported : YES
```

```
H14-2 (Deltavar_treatment > Deltavar_control):
Mean delta treatment = 0.0262 ± 0.1884
Mean delta control = 0.0064 ± 0.1780
Difference (T - C) = 0.0198
t = -3.9599, p (one-sided) = 0.0000
Supported : YES
```

3.4.2 H15

3.4.2.1 H15. Higher relative social circle variance decreases issue weight.

We model the new issue weights, $\alpha_{n,q}(w_2)$, inferred in wave 2, of the participants in the treatment condition as

$$\alpha_{n,q}(w_2) = \alpha_{n,q}(w_1) + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \frac{\text{var}_{n,q}(w_2) - \text{var}_{n,q}(w_1)}{\sum_q \text{var}_{n,q}(w_2) - \text{var}_{n,q}(w_1)} + \epsilon_{n,q} \quad (19)$$

The prediction is $\beta_1 < 0$. Higher relative social circle variance on one issue decreases the weight of that issue for the participant. %We test this via a t-test with significance threshold $\alpha = 0.05$.

UPDATE: There should be abs signs in the denominator! Otherwise small delta Var's are going to blow up things dramatically.

UPDATE: This does not need to depend on the Treatment condition group alone. I analysed both now

OPEN QUESTION: Is the denominator: $\sum_q |\text{var}_{n,q}(w_2) - \text{var}_{n,q}(w_1)|$ or $\sum_q |\text{var}_{n,q}(w_1)|$?? I chose the first now because this is what we preregistered.

```
variance_type <- "std"
k <- "exp"
var_col <- paste0(variance_type, "_socialCircle_ops")
```

```
# Variance change per participant per issue
df_h15_var <- df_p |>
  pivot_longer(cols = matches(paste0(var_col, "_")),
               names_to = c(".value", "issue"),
               names_pattern = paste0("^(", var_col, ")_(.*)")) |>
  rename(var_sc = all_of(var_col)) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
              names_prefix = "w", values_from = var_sc) |>
  rename(var_w1 = w1, var_w2 = w2) |>
  mutate(delta_var = var_w2 - var_w1) |>
  group_by(id) |>
  mutate(
    sum_delta_var = sum(abs(delta_var), na.rm = TRUE),
    # denominator is  $E|Deltavar|$  (preregistered); see note in LaTeX about alternatives
    rel_delta_var = if_else(sum_delta_var > 0.05, delta_var / sum_delta_var, NA_real_)
  ) |>
  ungroup()
```

```
# Alpha weights across waves
df_h15_alpha <- df_p |>
  pivot_longer(cols = matches(paste0("^", k, "_alpha_")),
               names_to = c(".value", "issue"),
               names_pattern = paste0("^(", k, "_alpha)_(.*)")) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
              names_prefix = "alpha_w", values_from = all_of(paste0(k, "_alpha")))
```

```
df_h15 <- df_h15_var |>
  left_join(df_h15_alpha, by = c("id", "issue")) |>
  left_join(df_p |> filter(wave == 2) |> select(id, treatment_wave2), by = "id")
```

```
m_h15 <- lm(alpha_w2 ~ alpha_w1 + rel_delta_var, data = df_h15)
cf_h15 <- summary(m_h15)$coefficients["rel_delta_var", ]
```

```
print_one_sided(
  sprintf("H15: Relative Delta social circle %s decreases issue weight (%s)", variance_type, k),
  cf_h15["Estimate"], cf_h15["Std. Error"], cf_h15["t value"],
  df = m_h15$df.residual,
  direction = "negative"
)
```

```
=== H15: Relative Delta social circle std decreases issue weight (exp) ===
```

```
Estimate = -0.0595
SE        = 0.0143
t         = -4.1464
df        = 5373.0
p (one-sided, beta < 0) = 0.0000
Supported : YES
```

```
df_h15 |>
  filter(!is.na(rel_delta_var), !is.na(alpha_w2), !is.na(alpha_w1)) |>
  mutate(
    delta_alpha = alpha_w2 - alpha_w1,
    rel_delta_var_bin = cut(rel_delta_var, breaks = 7),
    treatment_wave2 = factor(treatment_wave2, levels = c(0, 1),
                             labels = c("Control", "Treatment"))
  ) |>
  ggplot(aes(x = rel_delta_var, y = delta_alpha, color = treatment_wave2)) +
  geom_boxplot(aes(group = rel_delta_var_bin), fill = "steelblue", alpha = 0.15,
               outlier.shape = NA, color = "gray40") +
  geom_point(alpha = 0.1, size = 1) +
  geom_smooth(formula = y ~ x, method = "lm", se = FALSE, alpha = 0.5) +
  geom_smooth(formula = y ~ x, method = "loess", se = FALSE, linetype = "dashed", alpha = 0.5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  scale_color_manual(values = c("Control" = "steelblue", "Treatment" = "firebrick"),
```

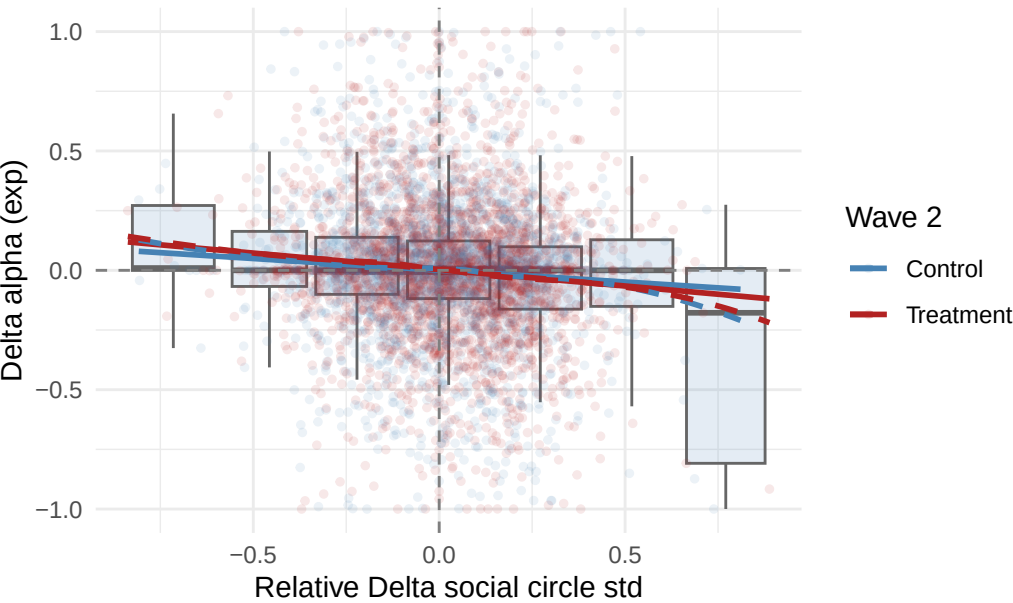
```

name = "Wave 2") +
labs(title = sprintf("H15: Relative social-circle-%s change vs weight (%s) change",
  variance_type, k),
x = sprintf("Relative Delta social circle %s", variance_type),
y = sprintf("Delta alpha (%s)", k)) +
theme_minimal()

```

Warning: Orientation is not uniquely specified when both the x and y aesthetics are continuous. Picking default orientation 'x'.

H15: Relative social-circle-std change vs weight (exp) change



3.4.3 H16 (caution)

3.4.3.1 H16. Increase in overall social circle variance decreases distance to typical voters.

We model the map distances between the participant themselves and typical voters of political parties r in wave w as

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \sum_q \Delta x_{n,self-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + \epsilon_{n,r} \quad (20)$$

The explanatory variable is the change in social circle variance averaged over all six issues and weighted by the corresponding belief differences between participant and the typical voter of party r , $\Delta x_{n,self-r,q} = |x_{n,self,q} - x_{n,r,q}|$.

The prediction is $\beta_1 < 0$. When overall social circle variance increases, the reported distance between participant and typical voter beliefs decreases. We test this via a t-test with significance threshold $\alpha = 0.05$.

ERROR: This should not include the treatment. The effect we want to test is whether variance increase (weighted by opinion difference, such that) changes distances.

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot \sum_q \Delta x_{n,self-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + \beta_2 \cdot \text{condition}_n + u_n + \epsilon_{n,r} \quad (21)$$

or alternatively, but that's not really a fair thing to test and not really related to our theory (see discussion with Mirta):

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot \text{condition}_n + \epsilon_{n,r} \quad (22)$$

```
# Build shared data objects used by both H16 and H17 variants
```

```
variance_type <- "std"
var_col <- paste0(variance_type, "_socialCircle_ops")
```

```
# Wave 1 and 2 self-party distances
```

```
df_self_party_w1 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 1) |>
  select(id, dot2, pixel_dist_w1 = pixel_dist)
```

```
df_self_party_w2 <- df_diff |>
```

```

filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
select(id, dot2, pixel_dist_w2 = pixel_dist, sympathy_w2 = sympathy)

# deltaX (wave 2)
df_deltaX <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, dot2, all_of(col_deltas))

# Variance change per participant per issue (wide)
df_delta_var_wide <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0(var_col, "_")),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, ")_(.*)")) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
    names_prefix = "var_w", values_from = all_of(var_col)) |>
  mutate(delta_var = var_w2 - var_w1) |>
  pivot_wider(id_cols = id, names_from = issue,
    names_prefix = "delta_var_", values_from = delta_var)

# Long delta_var (for new H16/H17 moderation models)
df_delta_var_long <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0(var_col, "_")),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, ")_(.*)")) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
    names_prefix = "var_w", values_from = all_of(var_col)) |>
  mutate(delta_var = var_w2 - var_w1,
    delta_var_c = as.numeric(scale(delta_var, center = TRUE, scale = FALSE))) |>
  select(id, issue, delta_var, delta_var_c)

```

```

# Simple direct-effect check (does treatment affect pixel_dist?)
df_h16_simple <- df_self_party_w2 |>
  mutate(group = if_else(id %in% treatment_ids, "treatment", "control")) |>
  left_join(df_self_party_w1, by = c("id", "dot2"))

```

```

m_h16_direct <- lmer(pixel_dist_w2 ~ group + (1 | id) + (1 | dot2),
  data = df_h16_simple, REML = TRUE)
cat("=== H16 (simple: treatment → pixel_dist) ===\n")
print(summary(m_h16_direct)$coefficients)

```

```

# Full model: weighted variance change
df_h16_full <- df_deltaX |>
  left_join(df_self_party_w1, by = c("id", "dot2")) |>
  left_join(df_self_party_w2, by = c("id", "dot2")) |>
  left_join(df_delta_var_wide, by = "id") |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
    weighted_var_change = rowSums(
      across(all_of(col_deltas)) * across(all_of(paste0("delta_var_", questions_sc))),
      na.rm = TRUE
    )
  )

```

```

m_h16_full <- lmer(pixel_dist_w2 ~ pixel_dist_w1 + weighted_var_change + group + (1 | id),
  data = df_h16_full, REML = TRUE)
cf_h16 <- lmer_coef(m_h16_full, "weighted_var_change")

cat("\n")
print_one_sided("H16 (full): weighted Deltavar decreases perceived distance",
  cf_h16$beta, cf_h16$se, cf_h16$t, direction = "negative")

```

```

=== H16 (simple: treatment → pixel_dist) ===
      Estimate Std. Error      df    t value    Pr(>|t|)

```

(Intercept)	0.338015954	0.017448847	7.643035	19.3718221	8.988148e-08
grouptreatment	-0.006245272	0.008341309	900.000072	-0.7487161	4.542240e-01

```

=== H16 (full): weighted Deltavar decreases perceived distance ===
  Estimate   =    0.0082
    SE      =    0.0046
     t      =    1.7734
    df      = infty
  p (one-sided, beta < 0) = 0.9619
Supported : NO

```

3.4.4 H17 (caution)

3.4.4.1 H17. Increase in overall social circle variance increases likeability rating of typical voters.

Similarly, we model how participant n rates the likeability rating of typical voters of party r in wave 2 as

$$y_{n,r} = \beta_0 + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \sum_q \Delta x_{n,\text{self}-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + u_n + \epsilon_{n,r} \tag{23}$$

The prediction is $\beta_1 > 0$. When overall social circle variance increases, the likeability rating of typical voters of the political parties increases. %We test this via a t-test with significance threshold $\alpha = 0.05$.

Most simple: Does treatment affect liking? No obviously not!

```

# Simple direct-effect check
df_h17_simple <- df_self_party_w2 |>
  mutate(group = if_else(id %in% treatment_ids, "treatment", "control"))

m_h17_direct <- lmer(sympathy_w2 ~ group + (1 | id) + (1 | dot2),
                    data = df_h17_simple, REML = TRUE)
cat("=== H17 (simple: treatment → sympathy) ===\n")
print(summary(m_h17_direct)$coefficients)

# Full model
df_h17_full <- df_deltaX |>
  left_join(df_self_party_w2, by = c("id", "dot2")) |>
  left_join(df_delta_var_wide, by = "id") |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
    weighted_var_change = rowSums(
      across(all_of(col_deltas)) * across(all_of(paste0("delta_var_", questions_sc))),
      na.rm = TRUE
    )
  )

m_h17_full <- lmer(sympathy_w2 ~ weighted_var_change + group + (1 | id),
                  data = df_h17_full, REML = TRUE)
cf_h17 <- lmer_coef(m_h17_full, "weighted_var_change")

cat("\n")
print_one_sided("H17 (full): weighted Deltavar increases likeability",
                cf_h17$beta, cf_h17$se, cf_h17$t, direction = "positive")

```

```

=== H17 (simple: treatment → sympathy) ===

```

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.423891501	0.026777532	6.507399	15.8301182	1.939544e-06
grouptreatment	-0.003302603	0.008775184	899.999845	-0.3763571	7.067402e-01

```

=== H17 (full): weighted Deltavar increases likeability ===
  Estimate   =   -0.0302
    SE      =    0.0075
     t      =   -4.0329
    df      = infty
  p (one-sided, beta > 0) = 1.0000
Supported : NO

```


3.5 NEW: new versions of H16 and H17 (moderation)

3.5.1 H16 new

3.5.1.1 H16: Treatment increases social circle variance, which weakens the effect of opinion differences on perceived distance.

We test this hypothesis in two steps, conditional on H14 being confirmed (i.e., the nudge successfully increased social circle variance in the treatment group).

Step 1 — Intent-to-Treat (ITT): Does the treatment moderate the relationship between opinion differences $\Delta X_{n,r,q} = |x_{n,\text{self},q} - x_{n,r,q}|$ and perceived distance d ? We model:

$$d_{n,\text{self}-r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \mathbb{I}(\text{condition}_n = T) + \beta_3 \cdot \Delta X_{n,r,q} \cdot \mathbb{I}(\text{condition}_n = T) + u_n + v_r + v_q + \epsilon_{n,r,q} \tag{24}$$

where u_n is the participant random intercept, v_r the party voter random intercept, and v_q the question random intercept. The prediction is $\beta_3 < 0$: the treatment weakens the effect of opinion differences on perceived distance.

Step 2 — Variance as moderator: Does the actual change in variance $\Delta \text{var}_{n,q} = \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)$ moderate the $\Delta X \rightarrow d$ relationship, above and beyond treatment assignment?

$$d_{n,\text{self}-r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \Delta \text{var}_{n,q} + \beta_3 \cdot \Delta X_{n,r,q} \cdot \Delta \text{var}_{n,q} + u_n + v_r + v_q + \epsilon_{n,r,q} \tag{25}$$

The prediction is $\beta_3 < 0$: when social circle variance increases, the effect of opinion differences on perceived distance weakens. Step 2 speaks to the *mechanism*: it is the variance increase itself, not merely treatment assignment, that drives the moderation. Together, Steps 1 and 2 establish the causal chain:

$$\mathbb{I}(T) \xrightarrow{H14} \Delta \text{var} \xrightarrow{\beta_3 < 0} \text{weakens } (\Delta X \rightarrow d) \tag{26}$$

```
# Step 1 - ITT: does treatment moderate DeltaX -> distance?
df_h16_itt <- df_diff |>
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2,
         id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0("^(\deltaX|", var_col, ")_(", paste(questions_sc, collapse = "|"), ")$")),
               names_to = c(".value", "issue"),
               names_pattern = paste0("^(\deltaX|", var_col, ")_(.*)")) |>
  select(id, wave, dot2, pixel_dist, issue, deltaX, all_of(var_col)) |>
  mutate(treatment = as.integer(id %in% treatment_ids))

m_h16_itt <- lmer(pixel_dist ~ deltaX * treatment + (1 | id) + (1 | issue) + (1 | dot2),
                 data = df_h16_itt)
cf_itt_16 <- lmer_coef(m_h16_itt, "deltaX:treatment")

cat("\n")
print_one_sided("H16-new Step 1 (ITT): treatment weakens DeltaX -> distance",
               cf_itt_16$beta, cf_itt_16$se, cf_itt_16$t, direction = "negative")

# Step 2 - Deltavar as moderator
df_h16_var <- df_h16_itt |>
  left_join(df_delta_var_long, by = c("id", "issue"))

m_h16_var <- lmer(pixel_dist ~ deltaX * delta_var_c + (1 | id) + (1 | issue) + (1 | dot2),
                 data = df_h16_var)
cf_var_16 <- lmer_coef(m_h16_var, "deltaX:delta_var_c")

print_one_sided("H16-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX -> distance",
               cf_var_16$beta, cf_var_16$se, cf_var_16$t, direction = "negative")
```

```
=== H16-new Step 1 (ITT): treatment weakens DeltaX -> distance ===
Estimate = 0.0044
SE       = 0.0030
t        = 1.4548
df       = infty
p (one-sided, beta < 0) = 0.9271
Supported : NO

=== H16-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX -> distance ===
Estimate = 0.0035
SE       = 0.0071
```



```

t           = 0.4890
df          = infty
p (one-sided, beta < 0) = 0.6876
Supported : NO

```

3.5.2 H17 new

3.5.2.1 H17: Treatment increases social circle variance, which increases likeability ratings of typical party voters.

We test this hypothesis in two steps, conditional on H14 being confirmed (i.e., the nudge successfully increased social circle variance in the treatment group).

Step 1 — Intent-to-Treat (ITT): Does the treatment moderate the relationship between opinion differences $\Delta X_{n,r,q}$ and likeability ratings $y_{n,r}$? We model:

$$y_{n,r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \mathbb{I}(\text{condition}_n = T) + \beta_3 \cdot \Delta X_{n,r,q} \cdot \mathbb{I}(\text{condition}_n = T) + u_n + v_r + v_q + \epsilon_{n,r,q} \quad (27)$$

where u_n is the participant random intercept, v_r the party voter random intercept, and v_q the question random intercept. The prediction is $\beta_3 > 0$: the treatment weakens the negative effect of opinion differences on likeability, i.e. participants in the treatment condition rate typical voters of parties they disagree with more favourably.

Step 2 — Variance as moderator: Does the actual change in variance $\Delta \text{var}_{n,q} = \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)$ moderate the $\Delta X \rightarrow y$ relationship, above and beyond treatment assignment?

$$y_{n,r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \Delta \text{var}_{n,q} + \beta_3 \cdot \Delta X_{n,r,q} \cdot \Delta \text{var}_{n,q} + u_n + v_r + v_q + \epsilon_{n,r,q} \quad (28)$$

The prediction is $\beta_3 > 0$: when social circle variance increases, the negative effect of opinion differences on likeability weakens. Step 2 speaks to the *mechanism*: it is the variance increase itself, not merely treatment assignment, that drives the moderation. Together, Steps 1 and 2 establish the causal chain:

$$\mathbb{I}(T) \xrightarrow{H14} \Delta \text{var} \xrightarrow{\beta_3 > 0} \text{weakens } (\Delta X \rightarrow -y) \quad (29)$$

```

df_h17_itt <- df_diff |>
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2, # fixed: was "2"
    id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0("^(\deltaX|", var_col, ")_((", paste(questions_sc, collapse = "|"), ")$")),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(\deltaX|", var_col, ")_(.*)")) |>
  select(id, wave, dot2, sympathy, issue, deltaX, all_of(var_col)) |>
  mutate(treatment = as.integer(id %in% treatment_ids))

m_h17_itt <- lmer(sympathy ~ deltaX * treatment + (1 | id) + (1 | issue) + (1 | dot2),
  data = df_h17_itt)
cf_itt_17 <- lmer_coef(m_h17_itt, "deltaX:treatment")

cat("\n")
print_one_sided("H17-new Step 1 (ITT): treatment weakens DeltaX → low likeability",
  cf_itt_17$beta, cf_itt_17$se, cf_itt_17$t, direction = "positive")

df_h17_var <- df_h17_itt |>
  left_join(df_delta_var_long, by = c("id", "issue"))

m_h17_var <- lmer(sympathy ~ deltaX * delta_var_c + (1 | id) + (1 | issue) + (1 | dot2),
  data = df_h17_var)
cf_var_17 <- lmer_coef(m_h17_var, "deltaX:delta_var_c")

print_one_sided("H17-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX → low likeability",
  cf_var_17$beta, cf_var_17$se, cf_var_17$t, direction = "positive")

```

```

=== H17-new Step 1 (ITT): treatment weakens DeltaX → low likeability ===
Estimate = -0.0168
SE       = 0.0053
t        = -3.1633
df       = infty
p (one-sided, beta > 0) = 0.9992
Supported : NO

```

```

=== H17-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX → low likeability ===

```

```
Estimate  =  -0.0256
SE        =   0.0126
t         =  -2.0382
df        =  infty
p (one-sided, beta > 0) = 0.9792
Supported : NO

#
```

4 Exclusion criteria

Participants are excluded if they complete the survey in less than one third of the median completion time, provide invariant slider responses, or do not indicate their own beliefs on the six political issues.